MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49 PR RADIATOR SUPPORTING STRUCTURE

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-1595

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	26-10-2023	AS BUILT	SHD	SGC	PRI	
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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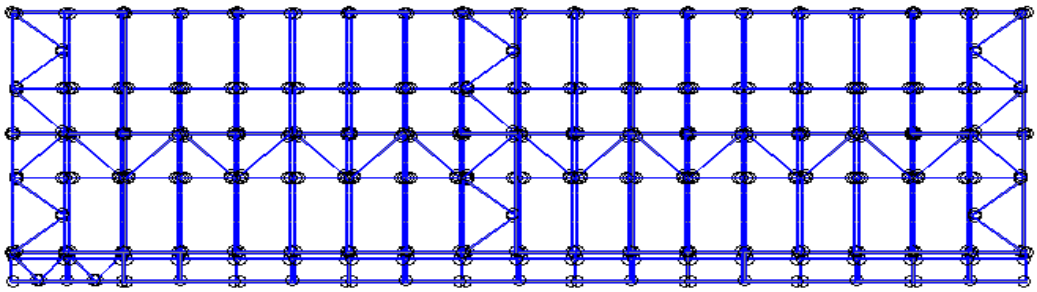
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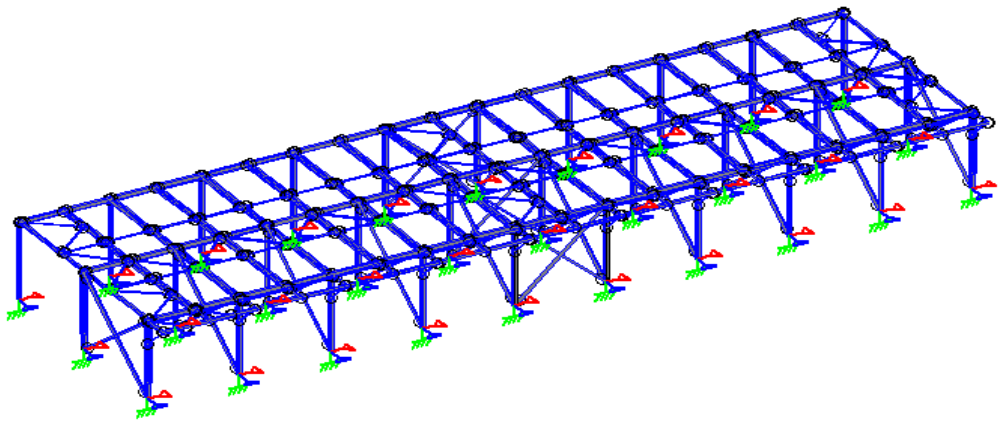
1. INTENT

The document is prepared for Design Calculation of Column Foundation for **Radiator Supporting Structure** of Power Plant.

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for Radiator supporting structure of Power Plant.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Foundations are kept 2.0m below FGL. Support Reactions from Superstructure have been applied as loads to Foundations.
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018
Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered





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3.0 DESIGN OF RADIATOR SUPPORTING STRUCTURE FOUNDATION

3.1 DESIGN OF FOUNDTION FOR COLUMN IN GRID "C"

FOUNDATION MKD F1	LOAD DATA FOR COL. C5/C6			(for most critical condition)
Load Detail	**Fx (T)**	**Fy(T)**	**Fz(T)**	
DEAD LOAD	0.1	1.3	0.0	
LIVE LOAD	0.0	0.0	0.0	
EQUIPMENT LOAD	0.5	3.5	0.0	
CABLE TRAY LOAD	0.3	2.4	0.0	
PIPE VERTICAL LOAD	0.0	0.0	0.0	
PIPE HORIZONTAL LOAD +X	-0.2	-0.5	0.0	
PIPE HORIZONTAL LOAD +Z	0.0	0.1	0.0	
TEMPERATURE LOAD +VE	2.8	0.1	0.0	
TEMPERATURE LOAD -VE	-2.8	-0.1	0.0	
WIND LOAD +X	3.3	6.0	0.0	
WIND LOAD -X	-3.3	-6.1	0.0	
WIND LOAD +Z	0.1	0.2	0.3	
WIND LOAD -Z	-0.1	-0.2	-0.3	
SEISMIC LOAD +X	0.8	1.5	0.0	
SEISMIC LOAD -X	-0.8	-1.5	0.0	
SEISMIC LOAD +Z	0.0	0.0	0.0	
SEISMIC LOAD -Z	0.0	0.0	0.0	

STRENGTH DESIGN	FOUNDATION IN C GRID			
Load Combinations	FX (T)	FY (T)	FZ (T)	
1.4D	1.04	9.60	-0.01	
1.4D+1.4(TL+ve)+1.7LL	4.9	9.8	0.0	**LOAD CONBINATION AS PER SBC 301 (2.3.2-2)**
1.4D+1.4(TL-ve)+1.7LL	-2.8	9.4	0.0	**LOAD CONBINATION AS PER SBC 301 (2.3.2-2)**
1.2D+1.2(TL+ve)+1.6LL	4.2	8.4	0.0	
1.2D+1.2(TL-ve)+1.6LL	-2.4	8.1	0.0	
1.2D + 1.6WL(+x)+0.5LL	6.2	17.9	0.0	**LOAD CONBINATION AS PER SBC 301 (2.3.2-1)**
1.2D + 1.6WL(-x)+0.5LL	-4.4	-1.4	0.0	
1.2D + 1.6WL(+z)+0.5LL	1.1	8.6	0.5	
1.2D + 1.6WL(-z)+0.5LL	0.7	7.9	-0.5	
1.2D + 1.0SL(+x)+0.5LL	1.7	9.7	0.0	
1.2D + 1.0SL(-x)+0.5LL	1.7	9.7	0.0	
1.2D + 1.0SL(+z)+0.5LL	0.9	8.3	0.0	
1.2D + 1.0SL(-z)+0.5LL	0.9	8.2	0.0	
0.9D+1.6WL(+x)	6.0	15.8	0.0	
0.9D+1.6WL(-x)	-4.6	-3.5	0.0	
0.9D+1.6WL(+z)	0.9	6.5	0.5	
0.9D+1.6WL(-z)	0.4	5.8	-0.5	
0.9D+SL(+x)	1.4	7.6	0.0	
0.9D+SL(-x)	-0.1	4.7	0.0	
0.9D+SL(+z)	0.7	6.2	0.0	
0.9D+SL(-z)	0.7	6.1	0.0	

MA'ADEN معادن Saudi Arabian Mining Company	Outotec	LARSEN & TOUBRO	DESIGN DOCUMENT Ma'aden Document Number: MD-216-49PR-EG-CV-CAL-1595 Outotec Document Number: 900707_GAU01_WG01_CED01_45013_OU100682339 L&T Document Number: O19045_008_C_K_24291_001

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DESIGN OF FOUNDATION MKD. F1

	About-X	About-Z
Size of column :	0.70m	0.70m
Size of fdn. raft :	2.25m	1.75m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-2.00 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

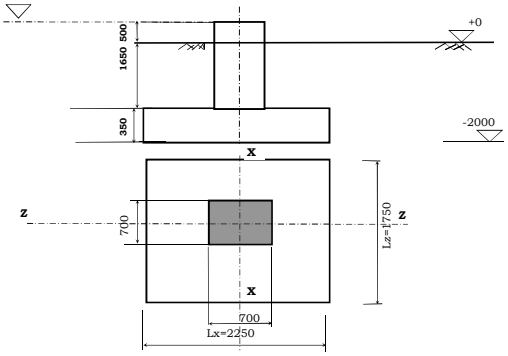
Dead Load	D
Temperature load	TL
Roof live load	Lr
Wind Load	WL
Seismic Load	SL

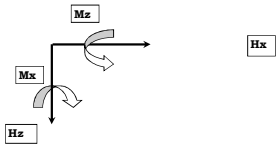
Calculation of Weight of Foundation :

Weight of raft & Col.= (2.25*1.75*0.35*2.5)+(0.7*0.7*2.15*2.5) = 6.08T

Weight of soil = ((2.25*1.75)-(0.7*0.7))*1.65*1.8 = 10.24T

U/S Base Pl Total = 16.32T





Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 267mm = 0.267m

Effective depth of slab along Z direction = 251mm = 0.251m

Clear projection along Z direction = 0.525m

Clear projection along X direction = 0.775m

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.75	6.87	-0.01	0.0	0.00	23.19	0.75	0.01	-0.01	1.50
DL+LL+TL(+ve)	3.52	7.01	-0.01	0.0	0.00	23.33	3.52	0.01	-0.01	7.04
DL+LL+TL(-ve)	-2.02	6.74	-0.01	0.0	0.00	23.06	-2.02	0.01	-0.01	4.05
DL+0.75*TL(+ve)+0.75*LL	2.83	6.97	-0.01	0.0	0.00	23.29	2.83	0.01	-0.01	5.65
DL+0.75*TL(-ve)+0.75*LL	-1.33	6.77	-0.01	0.0	0.00	23.09	-1.33	0.01	-0.01	2.67
DL+0.75*LL+0.75*WL(+x)	3.26	11.38	-0.01	0.0	0.00	27.70	3.26	0.01	-0.01	6.51
DL+0.75*LL+0.75*WL(-x)	-1.72	2.33	-0.01	0.0	0.00	18.65	-1.72	0.01	-0.01	3.44
DL+0.75*LL+0.75*WL(+z)	0.84	7.02	0.24	0.0	0.00	23.34	0.84	0.47	0.24	1.69
DL+0.75*LL+0.75*WL(-z)	0.64	6.71	-0.25	0.0	0.00	23.03	0.64	0.50	-0.25	1.29
DL+0.75*LL+0.75(0.7SL(+x))	1.15	7.96	-0.01	0.0	0.00	24.28	1.15	0.01	-0.01	2.30
DL+0.75*LL+0.75(0.7SL(-x))	0.35	5.77	-0.01	0.0	0.00	22.09	0.35	0.01	-0.01	0.69
DL+0.75*LL+0.75(0.7SL(+z))	0.76	6.89	-0.01	0.0	0.00	23.21	0.76	0.01	-0.01	1.51
DL+0.75*LL+0.75(0.7SL(-z))	0.74	6.85	-0.01	0.0	0.00	23.17	0.74	0.01	-0.01	1.48
0.6*DL+WL(+x)	3.79	10.13	0.00	0.0	0.00	26.45	3.79	0.01	0.00	7.58
0.6*DL+WL(-x)	-2.84	-1.94	0.00	0.0	0.00	14.38	-2.84	0.01	0.00	5.68
0.6*DL+WL(+z)	0.57	4.32	0.32	0.0	0.00	20.64	0.57	0.64	0.32	1.15
0.6*DL+WL(-z)	0.31	3.90	-0.33	0.0	0.00	20.22	0.31	0.66	-0.33	0.62
0.6*DL+0.7*SL(+x)	0.98	5.13	0.00	0.0	0.00	21.45	0.98	0.01	0.00	1.96
0.6*DL+0.7*SL(-x)	-0.09	3.09	0.00	0.0	0.00	19.41	-0.09	0.01	0.00	0.18
0.6*DL+0.7*SL(+z)	0.46	4.13	0.00	0.0	0.00	20.45	0.46	0.01	0.00	0.91
0.6*DL+0.7*SL(-z)	0.43	4.10	0.00	0.0	0.00	20.42	0.43	0.01	0.00	0.87

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CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	23.19	0.75	1.50	-0.01	0.01	13.55	1829.68	12.39	1301.10
DL+LL+TL(+ve)	23.33	3.52	7.04	-0.01	0.01	2.90	1840.33	2.65	1308.68
DL+LL+TL(-ve)	23.06	-2.02	4.05	-0.01	0.01	4.98	1819.03	4.56	1293.53
DL+0.75*TL(+ve)+0.75*LL	23.29	2.83	5.65	-0.01	0.01	3.60	1837.35	3.30	1306.56
DL+0.75*TL(-ve)+0.75*LL	23.09	-1.33	2.67	-0.01	0.01	7.58	1821.37	6.93	1295.20
DL+0.75*LL+0.75*WL(+x)	27.70	3.26	6.51	-0.01	0.01	3.72	2185.47	3.40	1554.11
DL+0.75*LL+0.75*WL(-x)	18.65	-1.72	3.44	-0.01	0.01	4.75	1471.23	4.34	1046.21
DL+0.75*LL+0.75*WL(+z)	23.34	0.84	1.69	0.24	0.47	12.11	55.77	11.07	39.66
DL+0.75*LL+0.75*WL(-z)	23.03	0.64	1.29	-0.25	0.50	15.62	51.88	14.28	36.89
DL+0.75*LL+0.75(0.7SL(+x))	24.28	1.15	2.30	-0.01	0.01	9.25	1915.66	8.46	1362.24
DL+0.75*LL+0.75(0.7SL(-x))	22.09	0.35	0.69	-0.01	0.01	27.97	1743.07	25.57	1239.51
DL+0.75*LL+0.75(0.7SL(+z))	23.21	0.76	1.51	-0.01	0.01	13.44	1830.93	12.28	1302.00
DL+0.75*LL+0.75(0.7SL(-z))	23.17	0.74	1.48	-0.01	0.01	13.73	1827.79	12.55	1299.76
0.6*DL+WL(+x)	26.45	3.79	7.58	0.00	0.01	3.05	3478.12	2.79	2473.33
0.6*DL+WL(-x)	14.38	-2.84	5.68	0.00	0.01	2.21	1890.91	2.02	1344.65
0.6*DL+WL(+z)	20.64	0.57	1.15	0.32	0.64	15.77	36.38	14.42	25.87
0.6*DL+WL(-z)	20.22	0.31	0.62	-0.33	0.66	28.67	34.71	26.21	24.68
0.6*DL+0.7*SL(+x)	21.45	0.98	1.96	0.00	0.01	9.58	2821.00	8.76	2006.04
0.6*DL+0.7*SL(-x)	19.41	-0.09	0.18	0.00	0.01	93.75	2552.53	85.71	1815.13
0.6*DL+0.7*SL(+z)	20.45	0.46	0.91	0.00	0.01	19.62	2689.21	17.93	1912.32
0.6*DL+0.7*SL(-z)	20.42	0.43	0.87	0.00	0.01	20.63	2684.32	18.86	1908.85

ULTIMATE STRESS DESIGN	**Load at U/S Of Base Plt**					**Load at bottom of foundation**		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	1.04	9.60	-0.01	0.0	0.00	25.919	0.02	2.59
1.4D+1.4(TL+ve)+1.7LL	4.93	9.81	-0.01	0.0	0.00	26.135	0.02	12.33
1.4D+1.4(TL-ve)+1.7LL	-2.83	9.44	-0.01	0.0	0.00	25.757	0.02	7.08
1.2D+1.2(TL+ve)+1.6LL	4.23	8.42	-0.01	0.0	0.00	24.735	0.02	10.57
1.2D+1.2(TL-ve)+1.6LL	-2.43	8.09	-0.01	0.0	0.00	24.411	0.02	6.06
1.2D + 1.6WL(+x)+0.5LL	6.25	17.87	-0.01	0.0	0.00	34.185	0.02	15.62
1.2D + 1.6WL(-x)+0.5LL	-4.37	-1.45	-0.01	0.0	0.00	14.871	0.02	10.91
1.2D + 1.6WL(+z)+0.5LL	1.10	8.56	0.51	0.0	0.00	24.879	1.27	2.74
1.2D + 1.6WL(-z)+0.5LL	0.68	7.89	-0.53	0.0	0.00	24.211	1.31	1.69
1.2D + 1.0SL(+x)+0.5LL	1.66	9.69	-0.01	0.0	0.00	26.014	0.02	4.14
1.2D + 1.0SL(-x)+0.5LL	1.66	9.69	-0.01	0.0	0.00	26.014	0.02	4.14
1.2D + 1.0SL(+z)+0.5LL	0.91	8.26	-0.01	0.0	0.00	24.582	0.02	2.27
1.2D + 1.0SL(-z)+0.5LL	0.88	8.21	-0.01	0.0	0.00	24.529	0.02	2.19
0.9D+1.6WL(+x)	6.02	15.80	-0.01	0.0	0.00	32.120	0.02	15.05
0.9D+1.6WL(-x)	-4.59	-3.51	-0.01	0.0	0.00	12.806	0.02	11.48
0.9D+1.6WL(+z)	0.87	6.49	0.51	0.0	0.00	22.814	1.28	2.18
0.9D+1.6WL(-z)	0.45	5.83	-0.52	0.0	0.00	22.146	1.31	1.12
0.9D+SL(+x)	1.43	7.63	-0.01	0.0	0.00	23.949	0.02	3.58
0.9D+SL(-x)	-0.10	4.71	-0.01	0.0	0.00	21.032	0.02	0.24
0.9D+SL(+z)	0.68	6.20	-0.01	0.0	0.00	22.517	0.02	1.71
0.9D+SL(-z)	0.65	6.14	-0.01	0.0	0.00	22.464	0.02	1.63

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	A02	2	3
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Calculation of Base Pressure

Load Combination & Calculation of Base pressure

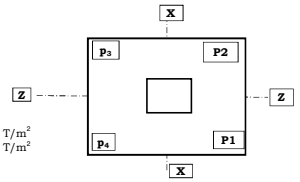
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.5 T/m^2 For WL/SL case = 67 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m^2) V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	Remarks
DL+LL	23.19	1.50	0.01	5.89	1.01	0.01	8.91	8.89	6.89	6.87	5.94	O.K
DL+LL+TL(+ve)	23.33	7.04	0.01	5.92	4.77	0.01	12.70	12.69	3.16	3.14	9.73	O.K
DL+LL+TL(-ve)	23.06	4.05	0.01	5.86	2.74	0.01	10.61	10.59	5.12	5.10	7.64	O.K
DL+0.75*TL(+ve)+0.75*LL	23.29	5.65	0.01	5.91	3.83	0.01	11.75	11.73	4.10	4.08	8.78	O.K
DL+0.75*TL(-ve)+0.75*LL	23.09	2.67	0.01	5.86	1.81	0.01	9.68	9.66	6.07	6.05	6.71	O.K
DL+0.75*LL+0.75*WL(+x)	27.70	6.51	0.01	7.04	4.41	0.01	13.46	13.44	4.63	4.61	10.49	O.K
DL+0.75*LL+0.75*WL(-x)	18.65	3.44	0.01	4.74	2.33	0.01	9.07	9.05	4.42	4.40	6.10	O.K
DL+0.75*LL+0.75*WL(+z)	23.34	1.69	0.47	5.93	1.14	0.32	9.39	8.75	7.10	6.47	6.42	O.K
DL+0.75*LL+0.75*WL(-z)	23.03	1.29	0.50	5.85	0.87	0.34	9.06	8.38	7.31	6.64	6.09	O.K
DL+0.75*LL+0.75(0.7SL(+x))	24.28	2.30	0.01	6.17	1.56	0.01	9.73	9.71	6.62	6.60	6.76	O.K
DL+0.75*LL+0.75(0.7SL(-x))	22.09	0.69	0.01	5.61	0.47	0.01	8.09	8.07	7.15	7.13	5.12	O.K
DL+0.75*LL+0.75(0.7SL(+z))	23.21	1.51	0.01	5.89	1.02	0.01	8.93	8.91	6.88	6.86	5.96	O.K
DL+0.75*LL+0.75(0.7SL(-z))	23.17	1.48	0.01	5.88	1.00	0.01	8.89	8.87	6.89	6.87	5.92	O.K
0.6*DL+WL(+x)	26.45	7.58	0.01	6.72	5.14	0.01	13.86	13.85	3.59	3.58	10.89	O.K
0.6*DL+WL(-x)	14.38	5.68	0.01	3.65	3.85	0.01	9.51	9.50	1.81	1.80	6.54	O.K
0.6*DL+WL(+z)	20.64	1.15	0.64	5.24	0.78	0.43	8.45	7.58	6.90	6.03	5.48	O.K
0.6*DL+WL(-z)	20.22	0.62	0.66	5.13	0.42	0.44	8.00	7.11	7.16	6.27	5.03	O.K
0.6*DL+0.7*SL(+x)	21.45	1.96	0.01	5.45	1.33	0.01	8.78	8.77	6.13	6.12	5.81	O.K
0.6*DL+0.7*SL(-x)	19.41	0.18	0.01	4.93	0.12	0.01	7.06	7.05	6.81	6.80	4.09	O.K
0.6*DL+0.7*SL(+z)	20.45	0.91	0.01	5.19	0.62	0.01	7.82	7.81	6.58	6.57	4.85	O.K
0.6*DL+0.7*SL(-z)	20.42	0.87	0.01	5.18	0.59	0.01	7.78	7.77	6.60	6.59	4.81	O.K

Calculation of Design Base Pressure

Load Combination & Calculation of Base pressure

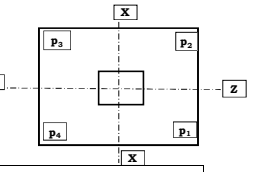
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.5 T/m^2 For WL/SL case = 67 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)
1.4D	25.9	2.59	0.02	6.58	1.76	0.02	10.36	10.32	6.84	6.81	6.76
1.4D+1.4(TL+ve)+1.7LL	26.1	12.33	0.02	6.64	8.35	0.02	17.01	14.97	-1.70	-1.73	13.41
1.4D+1.4(TL-ve)+1.7LL	25.8	7.08	0.02	6.54	4.79	0.02	13.35	11.32	1.76	1.73	9.75
1.2D+1.2(TL+ve)+1.6LL	24.7	10.57	0.02	6.28	7.16	0.01	15.46	13.43	-0.87	-0.89	11.86
1.2D+1.2(TL-ve)+1.6LL	24.4	6.06	0.02	6.20	4.11	0.01	12.32	10.29	2.11	2.08	8.72
1.2D + 1.6WL(+x)+0.5LL	34.2	15.62	0.02	8.68	10.58	0.01	21.27	19.25	-1.88	-1.91	17.67
1.2D + 1.6WL(-x)+0.5LL	14.9	10.91	0.02	3.78	7.39	0.01	13.18	11.15	-3.60	-3.63	9.58
1.2D + 1.6WL(+z)+0.5LL	24.9	2.74	1.27	6.32	1.86	0.86	11.04	7.32	5.32	3.60	7.44
1.2D + 1.6WL(-z)+0.5LL	24.2	1.69	1.31	6.15	1.14	0.89	10.18	6.40	5.90	4.11	6.58
1.2D + 1.0SL(+x)+0.5LL	26.0	4.14	0.02	6.61	2.81	0.01	11.43	9.40	3.81	3.79	7.83
1.2D + 1.0SL(-x)+0.5LL	26.0	4.14	0.02	6.61	2.81	0.01	11.43	9.40	3.81	3.79	7.83
1.2D + 1.0SL(+z)+0.5LL	24.6	2.27	0.02	6.24	1.54	0.01	9.80	7.77	4.72	4.69	6.20
1.2D + 1.0SL(-z)+0.5LL	24.5	2.19	0.02	6.23	1.48	0.01	9.73	7.70	4.76	4.73	6.13
0.9D+1.6WL(+x)	32.1	15.05	0.02	8.16	10.20	0.01	20.36	18.34	-2.03	-2.05	16.76
0.9D+1.6WL(-x)	12.8	11.48	0.02	3.25	7.77	0.01	13.04	11.02	-4.51	-4.53	9.44
0.9D+1.6WL(+z)	22.8	2.18	1.28	5.79	1.48	0.87	10.14	6.40	5.18	3.45	6.54
0.9D+1.6WL(-z)	22.1	1.12	1.31	5.62	0.76	0.89	9.27	5.50	5.75	3.98	5.67
0.9D+SL(+x)	23.9	3.58	0.02	6.08	2.42	0.01	10.52	8.49	3.67	3.65	6.92
0.9D+SL(-x)	21.0	0.24	0.02	5.34	0.17	0.01	7.52	5.50	5.19	5.17	3.92
0.9D+SL(+z)	22.5	1.71	0.02	5.72	1.16	0.01	8.89	6.86	4.57	4.55	5.29
0.9D+SL(-z)	22.5	1.63	0.02	5.71	1.10	0.01	8.82	6.80	4.61	4.59	5.22

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	A02	2	3
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.7+2*0.5*0.251)+(0.7+2*0.5*0.251))*2

= 3.804 m

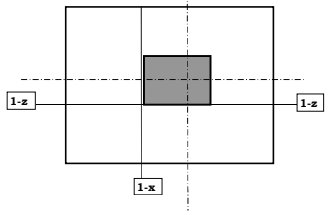
Punching Area= ((0.7+2*0.5*0.251)*(0.7+2*0.5*0.251))

= 0.9 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*2.5+1.65*1.8)

= 3.85 T/m^2

Area of footing, A_f= 3.94 m^2



Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 53.67 T

Strength of Concrete in shear Vc1 = 204.20 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 218.38 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 188.29 T **Table 22.6.5.2_ (c)**

Design Vc= 188.29 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (53.67*10000)/(3804*251)

= 0.56 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ ™	$S.F_{(oneway)}$ (T)	$B.M_{1z}$	$S.F_{(Oneway)}$ (T)	$S.F_{(punching)}$	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	6.76	0.93	3.43	2.03	1.85	20.53	0.22	OK	0.74	0.137	OK	0.32
1.4D+1.4(TL+ve)+1.7LL	13.41	1.85	6.81	4.03	3.67	40.73	0.43	OK	0.74	0.271	OK	0.64
1.4D+1.4(TL-ve)+1.7LL	9.75	1.34	4.95	2.93	2.67	29.62	0.31	OK	0.74	0.197	OK	0.46
1.2D+1.2(TL+ve)+1.6LL	11.86	1.63	6.02	3.56	3.25	36.02	0.38	OK	0.74	0.24	OK	0.57
1.2D+1.2(TL-ve)+1.6LL	8.72	1.20	4.43	2.62	2.39	26.49	0.28	OK	0.74	0.176	OK	0.42
1.2D + 1.6WL(+x)+0.5LL	17.67	2.44	8.98	5.31	4.84	53.67	0.56	OK	0.74	0.358	OK	0.84
1.2D + 1.6WL(-x)+0.5LL	9.58	1.32	4.87	2.88	2.62	29.10	0.3	OK	0.74	0.194	OK	0.46
1.2D + 1.6WL(+z)+0.5LL	7.44	1.03	3.78	2.23	2.04	22.60	0.24	OK	0.74	0.151	OK	0.35
1.2D + 1.6WL(-z)+0.5LL	6.58	0.91	3.34	1.98	1.80	19.99	0.21	OK	0.74	0.133	OK	0.31
1.2D + 1.0SL(+x)+0.5LL	7.83	1.08	3.98	2.35	2.15	23.78	0.25	OK	0.74	0.158	OK	0.37
1.2D + 1.0SL(-x)+0.5LL	7.83	1.08	3.98	2.35	2.15	23.78	0.25	OK	0.74	0.158	OK	0.37
1.2D + 1.0SL(+z)+0.5LL	6.20	0.85	3.15	1.86	1.70	18.83	0.2	OK	0.74	0.125	OK	0.30
1.2D + 1.0SL(-z)+0.5LL	6.13	0.84	3.11	1.84	1.68	18.62	0.2	OK	0.74	0.124	OK	0.29
0.9D+1.6WL(+x)	16.76	2.31	8.51	5.03	4.59	50.91	0.53	OK	0.74	0.339	OK	0.80
0.9D+1.6WL(-x)	9.44	1.30	4.80	2.83	2.59	28.67	0.3	OK	0.74	0.191	OK	0.45
0.9D+1.6WL(+z)	6.54	0.90	3.32	1.96	1.79	19.87	0.21	OK	0.74	0.132	OK	0.31
0.9D+1.6WL(-z)	5.67	0.78	2.88	1.70	1.55	17.22	0.18	OK	0.74	0.115	OK	0.27
0.9D+SL(+x)	6.92	0.95	3.52	2.08	1.90	21.02	0.22	OK	0.74	0.14	OK	0.33
0.9D+SL(-x)	3.92	0.54	1.99	1.18	1.07	11.91	0.12	OK	0.74	0.079	OK	0.19
0.9D+SL(+z)	5.29	0.73	2.69	1.59	1.45	16.07	0.17	OK	0.74	0.107	OK	0.25
0.9D+SL(-z)	5.22	0.72	2.65	1.57	1.43	15.86	0.17	OK	0.74	0.106	OK	0.25

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.43	0.001	267
1.4D+1.4(TL+ve)+1.7LL	0.85	0.002	534
1.4D+1.4(TL-ve)+1.7LL	0.62	0.001	267
1.2D+1.2(TL+ve)+1.6LL	0.75	0.002	534
1.2D+1.2(TL-ve)+1.6LL	0.55	0.001	267
1.2D + 1.6WL(+x)+0.5LL	1.12	0.003	801
1.2D + 1.6WL(-x)+0.5LL	0.61	0.001	267
1.2D + 1.6WL(+z)+0.5LL	0.47	0.001	267
1.2D + 1.6WL(-z)+0.5LL	0.42	0.001	267
1.2D + 1.0SL(+x)+0.5LL	0.5	0.001	267
1.2D + 1.0SL(-x)+0.5LL	0.5	0.001	267
1.2D + 1.0SL(+z)+0.5LL	0.39	0.001	267
1.2D + 1.0SL(-z)+0.5LL	0.39	0.001	267
0.9D+1.6WL(+x)	1.07	0.003	801
0.9D+1.6WL(-x)	0.6	0.001	267
0.9D+1.6WL(+z)	0.42	0.001	267
0.9D+1.6WL(-z)	0.36	0.001	267
0.9D+SL(+x)	0.44	0.001	267
0.9D+SL(-x)	0.25	0.001	267
0.9D+SL(+z)	0.34	0.001	267
0.9D+SL(-z)	0.33	0.001	267

Location	Mux (T-m)	ϕ Kn = M_{ux}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd.
At bottom	2.44	0.84	1.12	0.003	801.00	480.60	16φT	251.01	300	200
At top	0.00	0.00	0.00	0.0018	451.80	451.80	12φT	250.33	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction : At Bottom **16φT** **200** c/c At Top **12φT** **200** c/c

Along Z Direction : At Bottom **16φT** **200** c/c At Top **12φT** **200** c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
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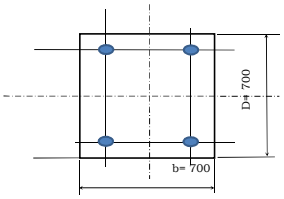
Outotec / L&T Revision No.:	0	A02	2	3
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
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DESIGN OF COLUMN MKD C1

Input:

Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	2150

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) =	2150.00
Unsupported length of column about Z (l_{uz}) =	2150.00

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio,	$K\,l_{ux}/r_x$ =	21.29	< 22 SECTION 6.2.5a
	$K\,l_{uz}/r_z$ =	21.29	< 22 SECTION 6.2.5a

Radius of gyration r_x= 202
Radius of gyration r_z= 202

γ_y = 0.75
γ_z = 0.75

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = $Pn/f_c'bD$ (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	1.04	9.6	0.0	3.4	14.8	0.0	0.01
1.4D+1.4(TL+ve)+1.7LL	4.93	9.8	0.0	16.3	15.1	0.0	0.01
1.4D+1.4(TL-ve)+1.7LL	-2.83	9.4	0.0	9.4	14.5	0.0	0.01
1.2D+1.2(TL+ve)+1.6LL	4.23	8.4	0.0	14.0	12.9	0.0	0.01
1.2D+1.2(TL-ve)+1.6LL	-2.43	8.1	0.0	8.0	12.4	0.0	0.01
1.2D + 1.6WL(+x)+0.5LL	6.25	17.9	0.0	20.7	27.5	0.0	0.02
1.2D + 1.6WL(-x)+0.5LL	-4.37	-1.4	0.0	14.4	-2.2	0.0	0.00
1.2D + 1.6WL(+z)+0.5LL	1.10	8.6	0.5	3.6	13.2	1.7	0.01
1.2D + 1.6WL(-z)+0.5LL	0.68	7.9	-0.5	2.2	12.1	1.7	0.01
1.2D + 1.0SL(+x)+0.5LL	1.66	9.7	0.0	5.5	14.9	0.0	0.01
1.2D + 1.0SL(-x)+0.5LL	1.66	9.7	0.0	5.5	14.9	0.0	0.01
1.2D + 1.0SL(+z)+0.5LL	0.91	8.3	0.0	3.0	12.7	0.0	0.01
1.2D + 1.0SL(-z)+0.5LL	0.88	8.2	0.0	2.9	12.6	0.0	0.01
0.9D+1.6WL(+x)	6.02	15.8	0.0	19.9	24.3	0.0	0.01
0.9D+1.6WL(-x)	-4.59	-3.5	0.0	15.2	-5.4	0.0	0.00
0.9D+1.6WL(+z)	0.87	6.5	0.5	2.9	10.0	1.7	0.01
0.9D+1.6WL(-z)	0.45	5.8	-0.5	1.5	9.0	1.7	0.01
0.9D+SL(+x)	1.43	7.6	0.0	4.7	11.7	0.0	0.01
0.9D+SL(-x)	-0.10	4.7	0.0	0.3	7.2	0.0	0.00
0.9D+SL(+z)	0.68	6.2	0.0	2.3	9.5	0.0	0.01
0.9D+SL(-z)	0.65	6.1	0.0	2.2	9.5	0.0	0.01

Global Project ID	**900707**
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Project Country	**KINGDOM OF SAUDI ARABIA**

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Provide,

4-Nos	25 mm Dia
4-Nos	25 mm Dia
2-Nos	25 mm Dia

1964.29 Sq mm (At Corner)
1964.29 Sq mm (At middle)
982.14 Sq mm (All round)

A_{sc} pvd. = 4910.714286

Total A_{sc} = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1649.39063 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cbD^2$	$R_{nx}=M_{nox}/f'_cbD^2$						
1.4D	0.06	0.06	72.0	72.03	0.55	1.16	0.03	O.K
1.4D+1.4(TL+ve)+1.7LL	0.06	0.06	72.0	72.03	0.55	1.16	0.18	O.K
1.4D+1.4(TL-ve)+1.7LL	0.06	0.06	72.0	72.03	0.55	1.16	0.09	O.K
1.2D+1.2(TL+ve)+1.6LL	0.06	0.06	72.0	72.03	0.55	1.16	0.15	O.K
1.2D+1.2(TL-ve)+1.6LL	0.06	0.06	72.0	72.03	0.55	1.16	0.08	O.K
1.2D + 1.6WL(+x)+0.5LL	0.07	0.07	84.0	84.04	0.55	1.16	0.20	O.K
1.2D + 1.6WL(-x)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.16	O.K
1.2D + 1.6WL(+z)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.04	O.K
1.2D + 1.6WL(-z)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.03	O.K
1.2D + 1.0SL(+x)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.05	O.K
1.2D + 1.0SL(-x)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.05	O.K
1.2D + 1.0SL(+z)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.03	O.K
1.2D + 1.0SL(-z)+0.5LL	0.06	0.06	72.0	72.03	0.55	1.16	0.02	O.K
0.9D+1.6WL(+x)	0.06	0.06	72.0	72.03	0.55	1.16	0.23	O.K
0.9D+1.6WL(-x)	0.06	0.06	72.0	72.03	0.55	1.16	0.16	O.K
0.9D+1.6WL(+z)	0.06	0.06	72.0	72.03	0.55	1.16	0.04	O.K
0.9D+1.6WL(-z)	0.06	0.06	72.0	72.03	0.55	1.16	0.02	O.K
0.9D+SL(+x)	0.06	0.06	72.0	72.03	0.55	1.16	0.04	O.K
0.9D+SL(-x)	0.06	0.06	72.0	72.03	0.55	1.16	0.00	O.K
0.9D+SL(+z)	0.06	0.06	72.0	72.03	0.55	1.16	0.02	O.K
0.9D+SL(-z)	0.06	0.06	72.0	72.03	0.55	1.16	0.02	O.K

Links, 10mm dia @ 300 C/C

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3.2 DESIGN OF FOUNDATION FOR COLUMN MKD B5,B6,A5,A6

FOUNDATION MKD F2	FOR COL B-6		
LOAD DEFINATION	Fx (T)	Fy (T)	Fz (T)
DEAD LOAD	-0.3	2.2	-0.1
LIVE LOAD	0.1	-0.4	0.2
EQUIPMENT LOAD	-1.2	7.1	-0.6
CABLE TRAY LOAD	0.0	0.0	-0.1
PIPE VERTICAL LOAD	0.0	0.0	-0.3
PIPE HORIZONTAL LOAD +X	1.3	-2.9	0.7
PIPE HORIZONTAL LOAD +Z	0.1	-1.3	0.9
TEMPERATURE LOAD +VE	-3.2	0.2	-4.1
TEMPERATURE LOAD -VE	3.2	-0.2	4.1
WIND LOAD +X	6.1	-10.9	0.6
WIND LOAD -X	-6.2	11.0	-0.6
WIND LOAD +Z	0.4	-4.5	3.1
WIND LOAD -Z	-0.4	4.5	-3.2
SEISMIC LOAD +X	1.2	-2.3	0.2
SEISMIC LOAD -X	-1.2	2.3	-0.2
SEISMIC LOAD +Z	0.1	-0.6	0.4
SEISMIC LOAD -Z	-0.1	0.6	-0.4

STRENGTH DESIGN	FOUNDATION AT B6		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	0.01	7.22	0.57
1.4D+1.4(TL+ve)+1.7LL	-4.25	6.79	-4.92
1.4D+1.4(TL-ve)+1.7LL	4.63	6.20	6.64
1.2D+1.2(TL+ve)+1.6LL	-3.63	5.76	-4.19
1.2D+1.2(TL-ve)+1.6LL	3.98	5.26	5.72
1.2D + 1.6WL(+x)+0.5LL	9.75	-11.53	1.58
1.2D + 1.6WL(-x)+0.5LL	-9.91	23.54	-0.44
1.2D + 1.6WL(+z)+0.5LL	0.72	-1.22	5.60
1.2D + 1.6WL(-z)+0.5LL	-0.57	13.18	-4.61
1.2D + 1.0SL(+x)+0.5LL	1.25	3.69	0.74
1.2D + 1.0SL(-x)+0.5LL	1.25	3.69	0.74
1.2D + 1.0SL(+z)+0.5LL	0.12	5.37	0.97
1.2D + 1.0SL(-z)+0.5LL	0.01	6.59	0.18
0.9D+1.6WL(+x)	9.70	-12.86	1.37
0.9D+1.6WL(-x)	-9.97	22.20	-0.65
0.9D+1.6WL(+z)	0.67	-2.55	5.39
0.9D+1.6WL(-z)	-0.62	11.84	-4.82
0.9D+SL(+x)	1.20	2.35	0.54
0.9D+SL(-x)	-1.18	6.93	0.19
0.9D+SL(+z)	0.06	4.03	0.76
0.9D+SL(-z)	-0.05	5.25	-0.03

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DESIGN OF FOUNDATION MKD.F2

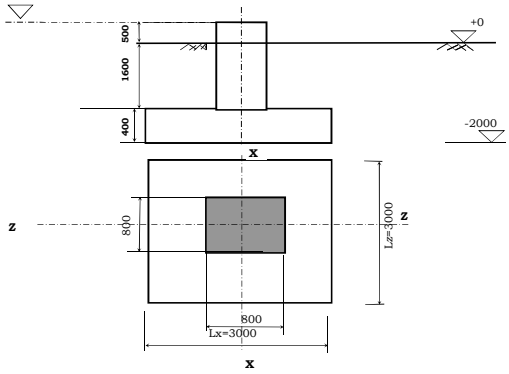
	About-X	About-Z
Size of column :	0.80m	0.80m
Size of fdn. raft :	3.00m	3.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-2.00 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

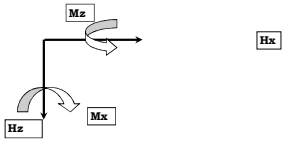
SYMBOLS AND NOTATIONS

Dead Load	D
Temperature load	TL
Roof live load	Lr
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(3*3*0.4*2.5)+(0.8*0.8*2.1*2.5) =	12.36T
Weight of soil =	((3*3)-(0.8*0.8))*1.6*1.8 =	24.08T
U/S Base Pl	Total =	36.44T





Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 317mm = 0.317m
Effective depth of slab along Z direction = 301mm = 0.301m
Clear projection along Z direction = 1.1m
Clear projection along X direction = 1.1m

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.11	4.73	0.58	0.0	0.00	41.17	0.11	1.45	0.58	0.28
DL+LL+TL(+ve)	-3.06	4.94	-3.55	0.0	0.00	41.38	-3.06	8.87	-3.55	7.65
DL+LL+TL(-ve)	3.28	4.52	4.71	0.0	0.00	40.96	3.28	11.77	4.71	8.21
DL+0.75*TL(+ve)+0.75*LL	-2.29	5.00	-2.56	0.0	0.00	41.44	-2.29	6.40	-2.56	5.73
DL+0.75*TL(-ve)+0.75*LL	2.47	4.68	3.63	0.0	0.00	41.12	2.47	9.08	3.63	6.16
DL+0.75*LL+0.75*WL(+x)	4.63	-3.37	1.01	0.0	0.00	33.07	4.63	2.52	1.01	11.57
DL+0.75*LL+0.75*WL(-x)	-4.59	13.07	0.06	0.0	0.00	49.51	-4.59	0.15	0.06	11.47
DL+0.75*LL+0.75*WL(+z)	0.40	1.47	2.89	0.0	0.00	37.91	0.40	7.23	2.89	0.99
DL+0.75*LL+0.75*WL(-z)	-0.21	8.21	-1.90	0.0	0.00	44.65	-0.21	4.74	-1.90	0.52
DL+0.75*LL+0.75(0.7SL(+x))	0.71	3.12	0.66	0.0	0.00	39.56	0.71	1.66	0.66	1.78
DL+0.75*LL+0.75(0.7SL(-x))	-0.54	6.56	0.41	0.0	0.00	43.00	-0.54	1.02	0.41	1.34
DL+0.75*LL+0.75(0.7SL(+z))	0.12	4.38	0.83	0.0	0.00	40.82	0.12	2.08	0.83	0.29
DL+0.75*LL+0.75(0.7SL(-z))	0.06	5.30	0.24	0.0	0.00	41.74	0.06	0.60	0.24	0.15
0.6*DL+WL(+x)	6.06	-7.84	0.87	0.0	0.00	28.60	6.06	2.18	0.87	15.15
0.6*DL+WL(-x)	-6.23	14.07	-0.39	0.0	0.00	50.51	-6.23	0.97	-0.39	15.57
0.6*DL+WL(+z)	0.42	-1.40	3.39	0.0	0.00	35.04	0.42	8.46	3.39	1.05
0.6*DL+WL(-z)	-0.39	7.59	-3.00	0.0	0.00	44.03	-0.39	7.50	-3.00	0.97
0.6*DL+0.7*SL(+x)	0.84	1.49	0.36	0.0	0.00	37.93	0.84	0.91	0.36	2.10
0.6*DL+0.7*SL(-x)	-0.83	4.70	0.12	0.0	0.00	41.14	-0.83	0.31	0.12	2.07
0.6*DL+0.7*SL(+z)	0.04	2.67	0.52	0.0	0.00	39.11	0.04	1.30	0.52	0.11
0.6*DL+0.7*SL(-z)	-0.03	3.52	-0.03	0.0	0.00	39.96	-0.03	0.08	-0.03	0.08

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Check for overturning and sliding:

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	41.17	0.11	0.28	0.58	1.45	218.97	42.69	145.98	28.46
DL+LL+TL(+ve)	41.38	-3.06	7.65	-3.55	8.87	8.12	6.99	5.41	4.66
DL+LL+TL(-ve)	40.96	3.28	8.21	4.71	11.77	7.48	5.22	4.99	3.48
DL+0.75*TL(+ve)+0.75*LL	41.44	-2.29	5.73	-2.56	6.40	10.85	9.71	7.23	6.47
DL+0.75*TL(-ve)+0.75*LL	41.12	2.47	6.16	3.63	9.08	10.01	6.79	6.67	4.53
DL+0.75*LL+0.75*WL(+x)	33.07	4.63	11.57	1.01	2.52	4.29	19.71	2.86	13.14
DL+0.75*LL+0.75*WL(-x)	49.51	-4.59	11.47	0.06	0.15	6.47	483.28	4.32	322.19
DL+0.75*LL+0.75*WL(+z)	37.91	0.40	0.99	2.89	7.23	57.28	7.86	38.19	5.24
DL+0.75*LL+0.75*WL(-z)	44.65	-0.21	0.52	-1.90	4.74	127.81	14.13	85.21	9.42
DL+0.75*LL+0.75(0.75L(+x))	39.56	0.71	1.78	0.66	1.66	33.37	35.80	22.25	23.87
DL+0.75*LL+0.75(0.75L(-x))	43.00	-0.54	1.34	0.41	1.02	48.01	63.25	32.01	42.17
DL+0.75*LL+0.75(0.75L(+z))	40.82	0.12	0.29	0.83	2.08	211.84	29.50	141.23	19.67
DL+0.75*LL+0.75(0.75L(-z))	41.74	0.06	0.15	0.24	0.60	428.95	104.08	285.96	69.38
0.6*DL+WL(+x)	28.60	6.06	15.15	0.87	2.18	2.83	19.68	1.89	13.12
0.6*DL+WL(-x)	50.51	-6.23	15.57	-0.39	0.97	4.87	77.99	3.24	52.00
0.6*DL+WL(+z)	35.04	0.42	1.05	3.39	8.46	50.16	6.21	33.44	4.14
0.6*DL+WL(-z)	44.03	-0.39	0.97	-3.00	7.50	67.79	8.81	45.19	5.87
0.6*DL+0.7*SL(+x)	37.93	0.84	2.10	0.36	0.91	27.15	62.80	18.10	41.86
0.6*DL+0.7*SL(-x)	41.14	-0.83	2.07	0.12	0.31	29.86	198.57	19.90	132.38
0.6*DL+0.7*SL(+z)	39.11	0.04	0.11	0.52	1.30	534.69	45.25	356.46	30.17
0.6*DL+0.7*SL(-z)	39.96	-0.03	0.08	-0.03	0.08	739.35	754.33	492.90	502.88

ULTIMATE STRESS DESIGN	Load at U/S Of Base Pit					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	0.01	7.22	0.57	0.0	0.00	43.661	1.42	0.03
1.4D+1.4(TL+ve)+1.7LL	-4.25	6.79	-4.92	0.0	0.00	43.230	12.29	10.63
1.4D+1.4(TL-ve)+1.7LL	4.63	6.20	6.64	0.0	0.00	42.645	16.61	11.57
1.2D+1.2(TL+ve)+1.6LL	-3.63	5.76	-4.19	0.0	0.00	42.199	10.48	9.07
1.2D+1.2(TL-ve)+1.6LL	3.98	5.26	5.72	0.0	0.00	41.698	14.29	9.96
1.2D + 1.6WL(+x)+0.5LL	9.75	-11.53	1.58	0.0	0.00	24.914	3.95	24.38
1.2D + 1.6WL(-x)+0.5LL	-9.91	23.54	-0.44	0.0	0.00	59.976	1.09	24.78
1.2D + 1.6WL(+z)+0.5LL	0.72	-1.22	5.60	0.0	0.00	35.223	14.00	1.81
1.2D + 1.6WL(-z)+0.5LL	-0.57	13.18	-4.61	0.0	0.00	49.615	11.54	1.42
1.2D + 1.0SL(+x)+0.5LL	1.25	3.69	0.74	0.0	0.00	40.125	1.86	3.13
1.2D + 1.0SL(-x)+0.5LL	1.25	3.69	0.74	0.0	0.00	40.125	1.86	3.13
1.2D + 1.0SL(+z)+0.5LL	0.12	5.37	0.97	0.0	0.00	41.806	2.42	0.29
1.2D + 1.0SL(-z)+0.5LL	0.01	6.59	0.18	0.0	0.00	43.027	0.45	0.02
0.9D+1.6WL(+x)	9.70	-12.86	1.37	0.0	0.00	23.579	3.43	24.25
0.9D+1.6WL(-x)	-9.97	22.20	-0.65	0.0	0.00	58.642	1.62	24.92
0.9D+1.6WL(+z)	0.67	-2.55	5.39	0.0	0.00	33.889	13.48	1.68
0.9D+1.6WL(-z)	-0.62	11.84	-4.82	0.0	0.00	48.281	12.06	1.56
0.9D+SL(+x)	1.20	2.35	0.54	0.0	0.00	38.791	1.34	2.99
0.9D+SL(-x)	-1.18	6.93	0.19	0.0	0.00	43.373	0.49	2.95
0.9D+SL(+z)	0.06	4.03	0.76	0.0	0.00	40.471	1.90	0.16
0.9D+SL(-z)	-0.05	5.25	-0.03	0.0	0.00	41.692	0.07	0.11

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	A02	2	3	
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023			Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI	

Calculation of Base Pressure

Load Combination & Calculation of Base pressure

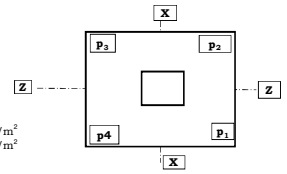
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.5 T/m^2 For WL/SL case = 67 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/(Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_{min})	Remarks
DL+LL	41.17	0.28	1.45	4.57	0.06	0.32	6.96	6.32	6.83	6.19	4.08	**O.K**
DL+LL+TL(+ve)	41.38	7.65	8.87	4.60	1.70	1.97	10.27	6.32	6.87	2.93	7.39	**O.K**
DL+LL+TL(-ve)	40.96	8.21	11.77	4.55	1.82	2.62	10.99	5.76	7.34	2.11	8.11	**O.K**
DL+0.75*TL(+ve)+0.75*LL	41.44	5.73	6.40	4.60	1.27	1.42	9.30	6.45	6.75	3.91	6.42	**O.K**
DL+0.75*TL(-ve)+0.75*LL	41.12	6.16	9.08	4.57	1.37	2.02	9.96	5.92	7.22	3.18	7.08	**O.K**
DL+0.75*LL+0.75*WL(+x)	33.07	11.57	2.52	3.67	2.57	0.56	8.81	7.69	3.66	2.54	5.93	**O.K**
DL+0.75*LL+0.75*WL(-x)	49.51	11.47	0.15	5.50	2.55	0.03	10.08	10.02	4.99	4.92	7.20	**O.K**
DL+0.75*LL+0.75*WL(+z)	37.91	0.99	7.23	4.21	0.22	1.61	8.04	4.83	7.60	4.38	5.16	**O.K**
DL+0.75*LL+0.75*WL(-z)	44.65	0.52	4.74	4.96	0.12	1.05	8.13	6.02	7.90	5.79	5.25	**O.K**
DL+0.75*LL+0.75(0.7SL(+x))	39.56	1.78	1.66	4.40	0.40	0.37	7.16	6.42	6.37	5.63	4.28	**O.K**
DL+0.75*LL+0.75(0.7SL(-x))	43.00	1.34	1.02	4.78	0.30	0.23	7.30	6.85	6.71	6.25	4.42	**O.K**
DL+0.75*LL+0.75(0.7SL(+z))	40.82	0.29	2.08	4.54	0.06	0.46	7.06	6.14	6.93	6.01	4.18	**O.K**
DL+0.75*LL+0.75(0.7SL(-z))	41.74	0.15	0.60	4.64	0.03	0.13	6.80	6.54	6.74	6.47	3.92	**O.K**
0.6*DL+WL(+x)	28.60	15.15	2.18	3.18	3.37	0.48	9.03	8.06	2.29	1.33	6.15	**O.K**
0.6*DL+WL(-x)	50.51	15.57	0.97	5.61	3.46	0.22	11.29	10.86	4.37	3.94	8.41	**O.K**
0.6*DL+WL(+z)	35.04	1.05	8.46	3.89	0.23	1.88	8.01	4.25	7.54	3.78	5.13	**O.K**
0.6*DL+WL(-z)	44.03	0.97	7.50	4.89	0.22	1.67	8.78	5.44	8.34	5.01	5.90	**O.K**
0.6*DL+0.7*SL(+x)	37.93	2.10	0.91	4.21	0.47	0.20	6.88	6.48	5.95	5.55	4.00	**O.K**
0.6*DL+0.7*SL(-x)	41.14	2.07	0.31	4.57	0.46	0.07	7.10	6.96	6.18	6.04	4.22	**O.K**
0.6*DL+0.7*SL(+z)	39.11	0.11	1.30	4.35	0.02	0.29	6.66	6.08	6.61	6.03	3.78	**O.K**
0.6*DL+0.7*SL(-z)	39.96	0.08	0.08	4.44	0.02	0.02	6.48	6.44	6.44	6.40	3.60	**O.K**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	A02	2	3
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023			Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI

CALCULATION OF DESIGN BASE PRESSURE

Load Combination & Calculation of Design Base pressure

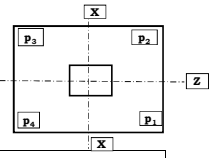
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.5 T/m^2 For WL/SL case = 67 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)							
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_3)
1.4D	43.7	0.03	1.42	4.85	0.01	0.32	7.17	6.54	7.16	6.53	3.57
1.4D+1.4(TL+ve)+1.7LL	43.2	10.63	12.29	4.80	2.36	2.73	11.90	6.43	7.17	1.71	8.30
1.4D+1.4(TL-ve)+1.7LL	42.6	11.57	16.61	4.74	2.57	3.69	13.00	5.62	7.86	0.48	9.40
1.2D+1.2(TL+ve)+1.6LL	42.2	9.07	10.48	4.69	2.02	2.33	11.03	6.38	7.00	2.34	7.43
1.2D+1.2(TL-ve)+1.6LL	41.7	9.96	14.29	4.63	2.21	3.18	12.02	5.67	7.60	1.24	8.42
1.2D + 1.6WL(+x)+0.5LL	24.9	24.38	3.95	2.77	5.42	0.88	11.06	9.31	0.23	-1.53	7.46
1.2D + 1.6WL(-x)+0.5LL	60.0	24.78	1.09	6.66	5.51	0.24	14.41	13.93	3.40	2.91	10.81
1.2D + 1.6WL(+z)+0.5LL	35.2	1.81	14.00	3.91	0.40	3.11	9.43	3.20	8.62	2.40	5.83
1.2D + 1.6WL(-z)+0.5LL	49.6	1.42	11.54	5.51	0.32	2.56	10.39	5.27	9.76	4.63	6.79
1.2D + 1.0SL(+x)+0.5LL	40.1	3.13	1.86	4.46	0.70	0.41	7.57	6.74	6.18	5.35	3.97
1.2D + 1.0SL(-x)+0.5LL	40.1	3.13	1.86	4.46	0.70	0.41	7.57	6.74	6.18	5.35	3.97
1.2D + 1.0SL(+z)+0.5LL	41.8	0.29	2.42	4.65	0.07	0.54	7.25	6.17	7.12	6.04	3.65
1.2D + 1.0SL(-z)+0.5LL	43.0	0.02	0.45	4.78	0.00	0.10	6.89	6.69	6.88	6.68	3.29
0.9D+1.6WL(+x)	23.6	24.25	3.43	2.62	5.39	0.76	10.77	9.25	-0.01	-1.53	7.17
0.9D+1.6WL(-x)	58.6	24.92	1.62	6.52	5.54	0.36	14.41	13.69	3.34	2.62	10.81
0.9D+1.6WL(+z)	33.9	1.68	13.48	3.77	0.37	3.00	9.13	3.14	8.39	2.40	5.53
0.9D+1.6WL(-z)	48.3	1.56	12.06	5.36	0.35	2.68	10.39	5.03	9.70	4.34	6.79
0.9D+SL(+x)	38.8	2.99	1.34	4.31	0.67	0.30	7.27	6.68	5.94	5.35	3.67
0.9D+SL(-x)	43.4	2.95	0.49	4.82	0.66	0.11	7.58	7.37	6.27	6.05	3.98
0.9D+SL(+z)	40.5	0.16	1.90	4.50	0.04	0.42	6.95	6.11	6.88	6.04	3.35
0.9D+SL(-z)	41.7	0.11	0.07	4.63	0.03	0.02	6.67	6.64	6.62	6.59	3.07

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM1z =Moment at the face of column 1z-1Z in T-m/m.

SF(oneway) = Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.7+2*0.5*0.251)+(0.7+2*0.5*0.251))*2

= 4.404m m

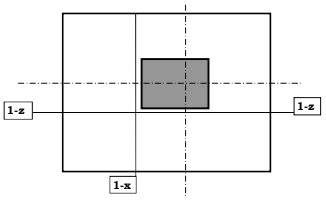
Punching Area= ((0.8+2*0.5*0.301)*(0.8+2*0.5*0.301))

= 1.21 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*2.5+1.6*1.8)

= 3.88 T/m^2

Area of footing, A_f= 9.00 m^2



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	A02	2	3
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 84.21 T

Strength of Concrete in shear Vc1 = 283.51 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 309.37 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 261.41 T **Table 22.6.5.2_ (c)**

Design Vc= 261.41 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (84.21*10000)/(4404*301)

= 0.64 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T/m^2)	B.M$_{1x}$ ™	S.F$_{(oneway)}$ (T)	B.M$_{1z}$ ™	S.F$_{(oneway)}$ (T)	S.F$_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	3.57	2.16	2.80	2.16	2.85	27.81	0.21	OK	0.74	0.095	OK	0.24
1.4D+1.4(TL+ve)+1.7LL	8.30	5.02	6.50	5.02	6.63	64.66	0.49	OK	0.74	0.22	OK	0.55
1.4D+1.4(TL-ve)+1.7LL	9.40	5.69	7.36	5.69	7.51	73.23	0.55	OK	0.74	0.25	OK	0.63
1.2D+1.2(TL+ve)+1.6LL	7.43	4.50	5.82	4.50	5.94	57.88	0.44	OK	0.74	0.197	OK	0.50
1.2D+1.2(TL-ve)+1.6LL	8.42	5.09	6.59	5.09	6.73	65.59	0.49	OK	0.74	0.224	OK	0.56
1.2D + 1.6WL(+x)+0.5LL	7.46	4.51	5.84	4.51	5.96	58.11	0.44	OK	0.74	0.198	OK	0.50
1.2D + 1.6WL(-x)+0.5LL	10.81	6.54	8.46	6.54	8.64	84.21	0.64	OK	0.74	0.287	OK	0.72
1.2D + 1.6WL(+z)+0.5LL	5.83	3.53	4.56	3.53	4.66	45.42	0.34	OK	0.74	0.155	OK	0.39
1.2D + 1.6WL(-z)+0.5LL	6.79	4.11	5.32	4.11	5.43	52.89	0.4	OK	0.74	0.18	OK	0.45
1.2D + 1.0SL(+x)+0.5LL	3.97	2.40	3.11	2.40	3.17	30.93	0.23	OK	0.74	0.105	OK	0.27
1.2D + 1.0SL(-x)+0.5LL	3.97	2.40	3.11	2.40	3.17	30.93	0.23	OK	0.74	0.105	OK	0.27
1.2D + 1.0SL(+z)+0.5LL	3.65	2.21	2.86	2.21	2.92	28.43	0.21	OK	0.74	0.097	OK	0.24
1.2D + 1.0SL(-z)+0.5LL	3.29	1.99	2.58	1.99	2.63	25.63	0.19	OK	0.74	0.087	OK	0.22
0.9D+1.6WL(+x)	7.17	4.34	5.61	4.34	5.73	55.85	0.42	OK	0.74	0.19	OK	0.48
0.9D+1.6WL(-x)	10.81	6.54	8.46	6.54	8.64	84.21	0.64	OK	0.74	0.287	OK	0.72
0.9D+1.6WL(+z)	5.53	3.35	4.33	3.35	4.42	43.08	0.32	OK	0.74	0.147	OK	0.37
0.9D+1.6WL(-z)	6.79	4.11	5.32	4.11	5.43	52.89	0.4	OK	0.74	0.18	OK	0.45
0.9D+SL(+x)	3.67	2.22	2.87	2.22	2.93	28.59	0.22	OK	0.74	0.097	OK	0.25
0.9D+SL(-x)	3.98	2.41	3.12	2.41	3.18	31.00	0.23	OK	0.74	0.106	OK	0.27
0.9D+SL(+z)	3.35	2.03	2.62	2.03	2.68	26.10	0.2	OK	0.74	0.089	OK	0.22
0.9D+SL(-z)	3.07	1.86	2.40	1.86	2.45	23.92	0.18	OK	0.74	0.081	OK	0.21

Load Case	kn (ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.32	0.001	317
1.4D+1.4(TL+ve)+1.7LL	0.74	0.002	634
1.4D+1.4(TL-ve)+1.7LL	0.84	0.002	634
1.2D+1.2(TL+ve)+1.6LL	0.66	0.002	634
1.2D+1.2(TL-ve)+1.6LL	0.75	0.002	634
1.2D + 1.6WL(+x)+0.5LL	0.66	0.002	634
1.2D + 1.6WL(-x)+0.5LL	0.96	0.002	634
1.2D + 1.6WL(+z)+0.5LL	0.52	0.001	317
1.2D + 1.6WL(-z)+0.5LL	0.6	0.001	317
1.2D + 1.0SL(+x)+0.5LL	0.35	0.001	317
1.2D + 1.0SL(-x)+0.5LL	0.35	0.001	317
1.2D + 1.0SL(+z)+0.5LL	0.32	0.001	317
1.2D + 1.0SL(-z)+0.5LL	0.29	0.001	317
0.9D+1.6WL(+x)	0.64	0.002	634
0.9D+1.6WL(-x)	0.96	0.002	634
0.9D+1.6WL(+z)	0.49	0.001	317
0.9D+1.6WL(-z)	0.6	0.001	317
0.9D+SL(+x)	0.33	0.001	317
0.9D+SL(-x)	0.35	0.001	317
0.9D+SL(+z)	0.3	0.001	317
0.9D+SL(-z)	0.27	0.001	317

Location	Mux (T-m)	ϕ Kn = M_{ux}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)
At bottom	6.54	0.72	0.96	0.002	634.00	570.60	16øT	317.13	300	200
At top	0.00	0.00	0.00	0.0018	541.80	541.80	12øT	208.74	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**16øT**	**200**	c/c	At Top	**12øT**	**200**	c/c
Along Z Direction :	At Bottom	**16øT**	**200**	c/c	At Top	**12øT**	**200**	c/c

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

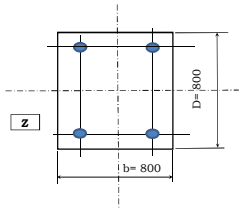
Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	D01 0	A02	2	3
Date	26-10-2023 07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	AS BUILT FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD SHD	SHD	SHD	SHD
Checked	SGC SGC	SGC	SGC	SGC
Approved	PRI PRI	PRI	PRI	PRI

DESIGN OF COLUMN MKD C2

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	2100

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) = 2100.00
Unsupported length of column about Z (l_{uz}) = 2100.00

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 18.19 < 22 SECTION 6.2.5a — Radius of gyration r_x= 231

$K\,l_{uz} / r_z$ = 18.19 < 22 SECTION 6.2.5a — Radius of gyration r_z= 231

γ_y =	0.78
γ_z =	0.78

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f_c'bD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	0.01	7.22	0.57	0.0	11.1	1.8	0.00
1.4D+1.4(TL+ve)+1.7LL	-4.25	6.79	-4.92	13.7	10.4	15.9	0.00
1.4D+1.4(TL-ve)+1.7LL	4.63	6.20	6.64	15.0	9.5	21.5	0.00
1.2D+1.2(TL+ve)+1.6LL	-3.63	5.76	-4.19	11.7	8.9	13.5	0.00
1.2D+1.2(TL-ve)+1.6LL	3.98	5.26	5.72	12.9	8.1	18.5	0.00
1.2D + 1.6WL(+x)+0.5LL	9.75	-11.53	1.58	31.5	-17.7	5.1	-0.01
1.2D + 1.6WL(-x)+0.5LL	-9.91	23.54	-0.44	32.0	36.2	1.4	0.02
1.2D + 1.6WL(+z)+0.5LL	0.72	-1.22	5.60	2.3	-1.9	18.1	0.00
1.2D + 1.6WL(-z)+0.5LL	-0.57	13.18	-4.61	1.8	20.3	14.9	0.01
1.2D + 1.0SL(+x)+0.5LL	1.25	3.69	0.74	4.0	5.7	2.4	0.00
1.2D + 1.0SL(-x)+0.5LL	1.25	3.69	0.74	4.0	5.7	2.4	0.00
1.2D + 1.0SL(+z)+0.5LL	0.12	5.37	0.97	0.4	8.3	3.1	0.00
1.2D + 1.0SL(-z)+0.5LL	0.01	6.59	0.18	0.0	10.1	0.6	0.00
0.9D+1.6WL(+x)	9.70	-12.86	1.37	31.3	-19.8	4.4	-0.01
0.9D+1.6WL(-x)	-9.97	22.20	-0.65	32.2	34.2	2.1	0.02
0.9D+1.6WL(+z)	0.67	-2.55	5.39	2.2	-3.9	17.4	0.00
0.9D+1.6WL(-z)	-0.62	11.84	-4.82	2.0	18.2	15.6	0.01
0.9D+SL(+x)	1.20	2.35	0.54	3.9	3.6	1.7	0.00
0.9D+SL(-x)	-1.18	6.93	0.19	3.8	10.7	0.6	0.00
0.9D+SL(+z)	0.06	4.03	0.76	0.2	6.2	2.4	0.00
0.9D+SL(-z)	-0.05	5.25	-0.03	0.1	8.1	0.1	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)	A_{sc} pvd. =	5892.86 Sq mm
4-Nos	25 mm Dia	1964.29 Sq mm	(At middle)		
4-Nos	25 mm Dia	1964.29 Sq mm	(At middle)	Total A_{sc} =	0.9 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 2133.96875 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cbD^2$						
1.4D	0.06	0.06	107.5	107.52	0.55	1.16	0.01	O.K
1.4D+1.4(TL+ve)+1.7LL	0.06	0.06	107.5	107.52	0.55	1.16	0.20	O.K
1.4D+1.4(TL-ve)+1.7LL	0.06	0.06	107.5	107.52	0.55	1.16	0.26	O.K
1.2D+1.2(TL+ve)+1.6LL	0.06	0.06	107.5	107.52	0.55	1.16	0.17	O.K
1.2D+1.2(TL-ve)+1.6LL	0.06	0.06	107.5	107.52	0.55	1.16	0.22	O.K
1.2D + 1.6WL(+x)+0.5LL	0.04	0.04	71.7	71.68	0.55	1.16	0.43	O.K
1.2D + 1.6WL(-x)+0.5LL	0.06	0.06	107.5	107.52	0.55	1.16	0.25	O.K
1.2D + 1.6WL(+z)+0.5LL	0.05	0.05	89.6	89.60	0.55	1.16	0.17	O.K
1.2D + 1.6WL(-z)+0.5LL	0.06	0.06	107.5	107.52	0.55	1.16	0.11	O.K
1.2D + 1.0SL(+x)+0.5LL	0.06	0.06	107.5	107.52	0.55	1.16	0.03	O.K
1.2D + 1.0SL(-x)+0.5LL	0.06	0.06	107.5	107.52	0.55	1.16	0.03	O.K
1.2D + 1.0SL(+z)+0.5LL	0.06	0.06	107.5	107.52	0.55	1.16	0.02	O.K
1.2D + 1.0SL(-z)+0.5LL	0.06	0.06	107.5	107.52	0.55	1.16	0.00	O.K
0.9D+1.6WL(+x)	0.04	0.04	71.7	71.68	0.55	1.16	0.42	O.K
0.9D+1.6WL(-x)	0.06	0.06	107.5	107.52	0.55	1.16	0.26	O.K
0.9D+1.6WL(+z)	0.06	0.06	107.5	107.52	0.55	1.16	0.13	O.K
0.9D+1.6WL(-z)	0.04	0.04	71.7	71.68	0.55	1.16	0.19	O.K
0.9D+SL(+x)	0.06	0.06	107.5	107.52	0.55	1.16	0.03	O.K
0.9D+SL(-x)	0.06	0.06	107.5	107.52	0.55	1.16	0.02	O.K
0.9D+SL(+z)	0.06	0.06	107.5	107.52	0.55	1.16	0.01	O.K
0.9D+SL(-z)	0.06	0.06	107.5	107.52	0.55	1.16	0.00	O.K

Links, **10mm dia @ 300 C/C**

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	A02	2	3
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
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3.3 DESIGN OF FOUNDATION FOR COLUMN MKD A1 TO A4, A7 TO A10, B1 TO B4, B7 TO B10

	LOAD DATA FOR COL. MKD B3		
Load Combinations	FX (T)	FY (T)	FZ (T)
DEAD LOAD	0.0	1.9	-0.2
LIVE LOAD	0.0	-0.4	0.1
EQUIPMENT LOAD	0.0	7.1	-0.8
CABLE TRAY LOAD	0.0	0.0	-0.2
PIPE VERTICAL LOAD	0.0	0.0	-0.4
PIPE HORIZONTAL LOAD +X	0.0	-0.1	0.0
PIPE HORIZONTAL LOAD +Z	0.0	-1.2	0.8
TEMPERATURE LOAD +VE	0.0	0.0	-4.5
TEMPERATURE LOAD -VE	0.0	0.0	4.5
WIND LOAD +X	0.4	0.0	0.0
WIND LOAD -X	-0.4	0.0	0.0
WIND LOAD +Z	0.0	-3.9	2.6
WIND LOAD -Z	0.0	3.9	-2.7
SEISMIC LOAD +X	0.0	0.0	0.0
SEISMIC LOAD -X	0.0	0.0	0.0
SEISMIC LOAD +Z	0.0	-0.6	0.4
SEISMIC LOAD -Z	0.0	0.6	-0.4

(for most critical condition)

Strength Design Load Combinations	LOAD DATA FOR COL. MKD B3		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	0.00	10.74	-0.91
1.4D+1.4(TL+ve)+1.7LL	0.00	9.98	-7.00
1.4D+1.4(TL-ve)+1.7LL	0.00	10.04	5.49
1.2D+1.2(TL+ve)+1.6LL	0.00	8.50	-5.98
1.2D+1.2(TL-ve)+1.6LL	0.00	8.55	4.72
1.2D + 1.6WL(+x)+0.5LL	0.66	9.05	-0.77
1.2D + 1.6WL(-x)+0.5LL	-0.66	9.00	-0.74
1.2D + 1.6WL(+z)+0.5LL	0.00	2.73	3.40
1.2D + 1.6WL(-z)+0.5LL	0.00	15.25	-5.02
1.2D + 1.0SL(+x)+0.5LL	0.00	9.00	-0.74
1.2D + 1.0SL(-x)+0.5LL	0.00	9.00	-0.74
1.2D + 1.0SL(+z)+0.5LL	0.00	8.41	-0.37
1.2D + 1.0SL(-z)+0.5LL	0.00	9.57	-1.11
0.9D+1.6WL(+x)	0.66	6.96	-0.62
0.9D+1.6WL(-x)	-0.66	6.91	-0.59
0.9D+1.6WL(+z)	0.00	0.65	3.55
0.9D+1.6WL(-z)	0.00	13.16	-4.87
0.9D+SL(+x)	0.00	6.91	-0.59
0.9D+SL(-x)	0.00	6.90	-0.58
0.9D+SL(+z)	0.00	6.33	-0.22
0.9D+SL(-z)	0.00	7.48	-0.96

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DESIGN OF FOUNDATION MKD. F3

	About-X	About-Z
Size of column :	0.70m	0.70m
Size of fdn. raft :	2.50m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-2.00 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

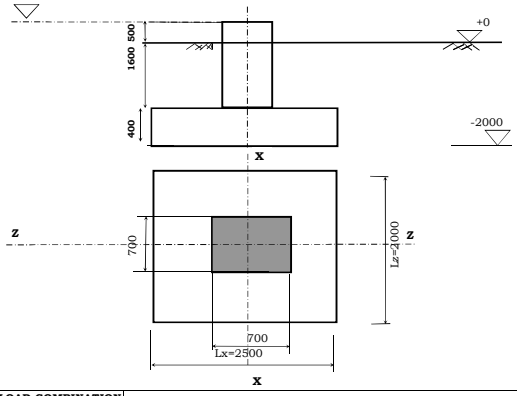
Dead Load	D
Temperature load	TL
Roof live load	Lr
Wind Load	WL
Seismic Load	SL

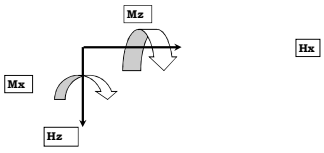
Calculation of Weight of Foundation :

Weight of raft & Col.= (2.5*2*0.4*2.5)+(0.7*0.7*2.1*2.5) = 7.57T

Weight of soil = ((2.5*2)-(0.7*0.7))*1.6*1.8 = 12.99T

U/S Base Pl Total = 20.56T





Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 315mm = 0.315m

Effective depth of slab along Z direction = 295mm = 0.295m

Clear projection along Z direction = 0.65m

Clear projection along X direction = 0.9m

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.00	7.24	-0.56	0.0	0.00	27.80	0.00	1.40	-0.56	0.00
DL+LL+TL(+ve)	0.00	7.22	-5.02	0.0	0.00	27.78	0.00	12.54	-5.02	0.00
DL+LL+TL(-ve)	0.00	7.26	3.90	0.0	0.00	27.82	0.00	9.75	3.90	0.00
DL+0.75*TL(+ve)+0.75*LL	0.00	7.33	-3.93	0.0	0.00	27.89	0.00	9.81	-3.93	0.00
DL+0.75*TL(-ve)+0.75*LL	0.00	7.37	2.76	0.0	0.00	27.93	0.00	6.90	2.76	0.00
DL+0.75*LL+0.75*WL(+x)	0.31	7.38	-0.60	0.0	0.00	27.94	0.31	1.50	-0.60	0.77
DL+0.75*LL+0.75*WL(-x)	-0.31	7.35	-0.58	0.0	0.00	27.91	-0.31	1.46	-0.58	0.77
DL+0.75*LL+0.75*WL(+z)	0.00	4.42	1.35	0.0	0.00	24.98	0.00	3.39	1.35	0.00
DL+0.75*LL+0.75*WL(-z)	0.00	10.28	-2.59	0.0	0.00	30.84	0.00	6.47	-2.59	0.00
DL+0.75*LL+0.75(0.75L(+x))	0.00	7.35	-0.58	0.0	0.00	27.91	0.00	1.46	-0.58	0.00
DL+0.75*LL+0.75(0.75L(-x))	0.00	7.35	-0.58	0.0	0.00	27.91	0.00	1.45	-0.58	0.00
DL+0.75*LL+0.75(0.75L(+z))	0.00	6.92	-0.31	0.0	0.00	27.48	0.00	0.76	-0.31	0.00
DL+0.75*LL+0.75(0.75L(-z))	0.00	7.78	-0.86	0.0	0.00	28.34	0.00	2.15	-0.86	0.00
0.6*DL+WL(+x)	0.41	4.64	-0.41	0.0	0.00	25.20	0.41	1.04	-0.41	1.03
0.6*DL+WL(-x)	-0.41	4.61	-0.39	0.0	0.00	25.17	-0.41	0.98	-0.39	1.03
0.6*DL+WL(+z)	0.00	0.69	2.19	0.0	0.00	21.25	0.00	5.48	2.19	0.00
0.6*DL+WL(-z)	0.00	8.51	-3.07	0.0	0.00	29.07	0.00	7.66	-3.07	0.00
0.6*DL+0.7*SL(+x)	0.00	4.61	-0.39	0.0	0.00	25.17	0.00	0.98	-0.39	0.00
0.6*DL+0.7*SL(-x)	0.00	4.60	-0.39	0.0	0.00	25.16	0.00	0.97	-0.39	0.00
0.6*DL+0.7*SL(+z)	0.00	4.20	-0.13	0.0	0.00	24.76	0.00	0.33	-0.13	0.00
0.6*DL+0.7*SL(-z)	0.00	5.01	-0.65	0.0	0.00	25.57	0.00	1.63	-0.65	0.00

Check for overturning and sliding:

	Load at Foundation Base					Overturning & Sliding Check (Coeffecient, μ = 0.4)			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+LL	27.80	0.00	0.00	-0.56	1.40	NA	24.86	NA	19.89
DL+LL+TL(+ve)	27.78	0.00	0.00	-5.02	12.54	NA	2.77	NA	2.22
DL+LL+TL(-ve)	27.82	0.00	0.00	3.90	9.75	NA	3.57	NA	2.86
DL+0.75*TL(+ve)+0.75*LL	27.89	0.00	0.00	-3.93	9.81	NA	3.55	NA	2.84
DL+0.75*TL(-ve)+0.75*LL	27.93	0.00	0.00	2.76	6.90	NA	5.06	NA	4.05
DL+0.75*LL+0.75*WL(+x)	27.94	0.31	0.77	-0.60	1.50	36.26	23.30	36.26	18.64
DL+0.75*LL+0.75*WL(-x)	27.91	-0.31	0.77	-0.58	1.46	36.23	23.90	36.23	19.12
DL+0.75*LL+0.75*WL(+z)	24.98	0.00	0.00	1.35	3.39	NA	9.22	NA	7.37
DL+0.75*LL+0.75*WL(-z)	30.84	0.00	0.00	-2.59	6.47	NA	5.96	NA	4.77
DL+0.75*LL+0.75(0.75L(+x))	27.91	0.00	0.00	-0.58	1.46	NA	23.86	NA	19.09
DL+0.75*LL+0.75(0.75L(-x))	27.91	0.00	0.00	-0.58	1.45	NA	24.05	NA	19.24
DL+0.75*LL+0.75(0.75L(+z))	27.48	0.00	0.00	-0.31	0.76	NA	44.95	NA	35.96
DL+0.75*LL+0.75(0.75L(-z))	28.34	0.00	0.00	-0.86	2.15	NA	16.49	NA	13.19
0.6*DL+WL(+x)	25.20	0.41	1.03	-0.41	1.04	24.53	30.41	24.53	24.33
0.6*DL+WL(-x)	25.17	-0.41	1.03	-0.39	0.98	24.50	31.99	24.50	25.59
0.6*DL+WL(+z)	21.25	0.00	0.00	2.19	5.48	NA	4.85	NA	3.88
0.6*DL+WL(-z)	29.07	0.00	0.00	-3.07	7.66	NA	4.74	NA	3.79
0.6*DL+0.7*SL(+x)	25.17	0.00	0.00	-0.39	0.98	NA	31.96	NA	25.57
0.6*DL+0.7*SL(-x)	25.16	0.00	0.00	-0.39	0.97	NA	32.31	NA	25.84
0.6*DL+0.7*SL(+z)	24.76	0.00	0.00	-0.13	0.33	NA	93.01	NA	74.41
0.6*DL+0.7*SL(-z)	25.57	0.00	0.00	-0.65	1.63	NA	19.67	NA	15.73

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	A02	2	3
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ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	0.00	10.74	-0.91	0.0	0.00	31.300	2.28	0.00
1.4D+1.4(TL+ve)+1.7LL	0.00	9.98	-7.00	0.0	0.00	30.542	17.49	0.00
1.4D+1.4(TL-ve)+1.7LL	0.00	10.04	5.49	0.0	0.00	30.601	13.71	0.00
1.2D+1.2(TL+ve)+1.6LL	0.00	8.50	-5.98	0.0	0.00	29.055	14.96	0.00
1.2D+1.2(TL-ve)+1.6LL	0.00	8.55	4.72	0.0	0.00	29.106	11.79	0.00
1.2D + 1.6WL(+x)+0.5LL	0.66	9.05	-0.77	0.0	0.00	29.612	1.93	1.64
1.2D + 1.6WL(-x)+0.5LL	-0.66	9.00	-0.74	0.0	0.00	29.556	1.85	1.64
1.2D + 1.6WL(+z)+0.5LL	0.00	2.73	3.40	0.0	0.00	23.293	8.49	0.00
1.2D + 1.6WL(-z)+0.5LL	0.00	15.25	-5.02	0.0	0.00	35.809	12.54	0.00
1.2D + 1.0SL(+x)+0.5LL	0.00	9.00	-0.74	0.0	0.00	29.557	1.85	0.00
1.2D + 1.0SL(-x)+0.5LL	0.00	9.00	-0.74	0.0	0.00	29.557	1.85	0.00
1.2D + 1.0SL(+z)+0.5LL	0.00	8.41	-0.37	0.0	0.00	28.973	0.92	0.00
1.2D + 1.0SL(-z)+0.5LL	0.00	9.57	-1.11	0.0	0.00	30.131	2.76	0.00
0.9D+1.6WL(+x)	0.66	6.96	-0.62	0.0	0.00	27.525	1.56	1.64
0.9D+1.6WL(-x)	-0.66	6.91	-0.59	0.0	0.00	27.469	1.48	1.64
0.9D+1.6WL(+z)	0.00	0.65	3.55	0.0	0.00	21.206	8.87	0.00
0.9D+1.6WL(-z)	0.00	13.16	-4.87	0.0	0.00	33.722	12.17	0.00
0.9D+SL(+x)	0.00	6.91	-0.59	0.0	0.00	27.470	1.48	0.00
0.9D+SL(-x)	0.00	6.90	-0.58	0.0	0.00	27.459	1.46	0.00
0.9D+SL(+z)	0.00	6.33	-0.22	0.0	0.00	26.885	0.55	0.00
0.9D+SL(-z)	0.00	7.48	-0.96	0.0	0.00	28.043	2.39	0.00

Calculation of Base Pressure

Load Combination & Calculation of Base pressure

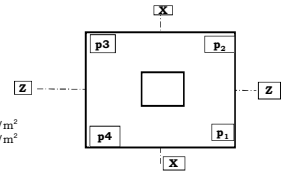
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.5 T/m^2 For WL/SL case = 67 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_{min})	Remarks
DL+LL	27.80	0.00	1.40	5.56	0.00	0.67	8.23	6.89	8.23	6.89	5.35	O.K
DL+LL+TL(+ve)	27.78	0.00	12.54	5.56	0.00	6.02	13.58	1.54	13.58	1.54	10.70	O.K
DL+LL+TL(-ve)	27.82	0.00	9.75	5.56	0.00	4.68	12.24	2.89	12.24	2.89	9.36	O.K
DL+0.75*TL(+ve)+0.75*LL	27.89	0.00	9.81	5.58	0.00	4.71	12.29	2.87	12.29	2.87	9.41	O.K
DL+0.75*TL(-ve)+0.75*LL	27.93	0.00	6.90	5.59	0.00	3.31	10.90	4.27	10.90	4.27	8.02	O.K
DL+0.75*LL+0.75*WL(+x)	27.94	0.77	1.50	5.59	0.46	0.72	8.77	7.33	7.85	6.41	5.89	O.K
DL+0.75*LL+0.75*WL(-x)	27.91	0.77	1.46	5.58	0.46	0.70	8.75	7.34	7.82	6.42	5.87	O.K
DL+0.75*LL+0.75*WL(+z)	24.98	0.00	3.39	5.00	0.00	1.63	8.62	5.37	8.62	5.37	5.74	O.K
DL+0.75*LL+0.75*WL(-z)	30.84	0.00	6.47	6.17	0.00	3.11	11.27	5.06	11.27	5.06	8.39	O.K
DL+0.75*LL+0.75(0.7SL(+x))	27.91	0.00	1.46	5.58	0.00	0.70	8.28	6.88	8.28	6.88	5.40	O.K
DL+0.75*LL+0.75(0.7SL(-x))	27.91	0.00	1.45	5.58	0.00	0.70	8.28	6.88	8.28	6.88	5.40	O.K
DL+0.75*LL+0.75(0.7SL(+z))	27.48	0.00	0.76	5.50	0.00	0.37	7.86	7.13	7.86	7.13	4.98	O.K
DL+0.75*LL+0.75(0.7SL(-z))	28.34	0.00	2.15	5.67	0.00	1.03	8.70	6.64	8.70	6.64	5.82	O.K
0.6*DL+WL(+x)	25.20	1.03	1.04	5.04	0.62	0.50	8.15	7.16	6.92	5.93	5.27	O.K
0.6*DL+WL(-x)	25.17	1.03	0.98	5.03	0.62	0.47	8.12	7.18	6.89	5.94	5.24	O.K
0.6*DL+WL(+z)	21.25	0.00	5.48	4.25	0.00	2.63	8.88	3.62	8.88	3.62	6.00	O.K
0.6*DL+WL(-z)	29.07	0.00	7.66	5.81	0.00	3.68	11.49	4.14	11.49	4.14	8.61	O.K
0.6*DL+0.7*SL(+x)	25.17	0.00	0.98	5.03	0.00	0.47	7.51	6.56	7.51	6.56	4.63	O.K
0.6*DL+0.7*SL(-x)	25.16	0.00	0.97	5.03	0.00	0.47	7.50	6.56	7.50	6.56	4.62	O.K
0.6*DL+0.7*SL(+z)	24.76	0.00	0.33	4.95	0.00	0.16	7.11	6.79	7.11	6.79	4.23	O.K
0.6*DL+0.7*SL(-z)	25.57	0.00	1.63	5.11	0.00	0.78	7.89	6.33	7.89	6.33	5.01	O.K

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	A02	2	3
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Calculation of Design Base Pressure

Load Combination & Calculation of Base pressure

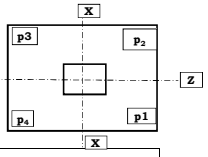
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 For WL/SL case = 62.5 T/m^2

Gross SBC of soil for DL, LL= 54.5 T/m^2 For WL/SL case = 67 T/m^2



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m^2) V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-p_{min})
1.4D	31.3	0.00	2.28	6.26	0.00	1.37	9.63	6.89	9.63	6.89	6.03
1.4D+1.4(TL+ve)+1.7LL	30.5	0.00	17.49	6.11	0.00	10.49	18.60	-2.38	18.60	-2.38	15.00
1.4D+1.4(TL-ve)+1.7LL	30.6	0.00	13.71	6.12	0.00	8.23	16.35	-0.11	16.35	-0.11	12.75
1.2D+1.2(TL+ve)+1.6LL	29.1	0.00	14.96	5.81	0.00	8.97	16.78	-1.16	16.78	-1.16	13.18
1.2D+1.2(TL-ve)+1.6LL	29.1	0.00	11.79	5.82	0.00	7.07	14.89	0.75	14.89	0.75	11.29
1.2D + 1.6WL(+x)+0.5LL	29.6	1.64	1.93	5.92	0.79	1.16	9.87	7.55	8.29	5.97	6.27
1.2D + 1.6WL(-x)+0.5LL	29.6	1.64	1.85	5.91	0.79	1.11	9.81	7.59	8.23	6.01	6.21
1.2D + 1.6WL(+z)+0.5LL	23.3	0.00	8.49	4.66	0.00	5.10	11.75	1.56	11.75	1.56	8.15
1.2D + 1.6WL(-z)+0.5LL	35.8	0.00	12.54	7.16	0.00	7.52	16.68	1.64	16.68	1.64	13.08
1.2D + 1.0SL(+x)+0.5LL	29.6	0.00	1.85	5.91	0.00	1.11	9.02	6.80	9.02	6.80	5.42
1.2D + 1.0SL(-x)+0.5LL	29.6	0.00	1.85	5.91	0.00	1.11	9.02	6.80	9.02	6.80	5.42
1.2D + 1.0SL(+z)+0.5LL	29.0	0.00	0.92	5.79	0.00	0.55	8.35	7.24	8.35	7.24	4.75
1.2D + 1.0SL(-z)+0.5LL	30.1	0.00	2.76	6.03	0.00	1.66	9.68	6.37	9.68	6.37	6.08
0.9D+1.6WL(+x)	27.5	1.64	1.56	5.50	0.79	0.94	9.23	7.36	7.65	5.78	5.63
0.9D+1.6WL(-x)	27.5	1.64	1.48	5.49	0.79	0.89	9.17	7.40	7.59	5.82	5.57
0.9D+1.6WL(+z)	21.2	0.00	8.87	4.24	0.00	5.32	11.56	0.92	11.56	0.92	7.96
0.9D+1.6WL(-z)	33.7	0.00	12.17	6.74	0.00	7.30	16.04	1.45	16.04	1.45	12.44
0.9D+SL(+x)	27.5	0.00	1.48	5.49	0.00	0.89	8.38	6.61	8.38	6.61	4.78
0.9D+SL(-x)	27.5	0.00	1.46	5.49	0.00	0.88	8.37	6.62	8.37	6.62	4.77
0.9D+SL(+z)	26.9	0.00	0.55	5.38	0.00	0.33	7.70	7.05	7.70	7.05	4.10
0.9D+SL(-z)	28.0	0.00	2.39	5.61	0.00	1.43	9.04	6.17	9.04	6.17	5.44

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

$SF_{(oneway)}$ = Oneway shear force at a distance 'd' from the face of column in T/m.

$SF_{(punching)}$ = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.7+2*0.5*0.251)+(0.7+2*0.5*0.251))*2

= 3.98 m

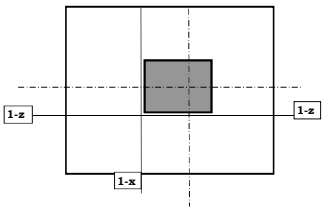
Punching Area= ((0.7+2*0.5*0.295)*(0.7+2*0.5*0.295))

= 0.99 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*2.5+1.6*1.8)

= 3.88 T/m^2

Area of footing, A_f= 5.00 m^2



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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 60.15 T

Strength of Concrete in shear Vc1 = 251.10 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 287.38 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 231.54 T **Table 22.6.5.2_ (c)**

Design Vc= 231.54 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (60.15*10000)/(3980*295)

= 0.51 N/mm^2 **No shear reinforcement reqd.**

Load Case	p_{1x} (T)	B.M$_{1x}$ (T-m)	S.F$_{(Oneway)}$ (T)	B.M$_{1z}$ T-M	S.F$_{(oneway)}$ (T)	S.F$_{(punching)}$ (T)	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	6.03	1.27	2.02	2.44	3.65	24.18	0.21	OK	0.74	0.124	OK	0.28
1.4D+1.4(TL+ve)+1.7LL	15.00	3.17	5.03	6.08	9.08	60.15	0.51	OK	0.74	0.308	OK	0.70
1.4D+1.4(TL-ve)+1.7LL	12.75	2.69	4.27	5.16	7.71	51.13	0.44	OK	0.74	0.261	OK	0.59
1.2D+1.2(TL+ve)+1.6LL	13.18	2.78	4.42	5.34	7.97	52.85	0.45	OK	0.74	0.27	OK	0.61
1.2D+1.2(TL-ve)+1.6LL	11.29	2.39	3.78	4.57	6.83	45.27	0.39	OK	0.74	0.232	OK	0.53
1.2D + 1.6WL(+x)+0.5LL	6.27	1.32	2.10	2.54	3.79	25.14	0.21	OK	0.74	0.129	OK	0.29
1.2D + 1.6WL(-x)+0.5LL	6.21	1.31	2.08	2.52	3.76	24.90	0.21	OK	0.74	0.127	OK	0.29
1.2D + 1.6WL(+z)+0.5LL	8.15	1.72	2.73	3.30	4.93	32.68	0.28	OK	0.74	0.167	OK	0.38
1.2D + 1.6WL(-z)+0.5LL	13.08	2.76	4.38	5.30	7.91	52.45	0.45	OK	0.74	0.268	OK	0.61
1.2D + 1.0SL(+x)+0.5LL	5.42	1.14	1.82	2.20	3.28	21.73	0.19	OK	0.74	0.111	OK	0.25
1.2D + 1.0SL(-x)+0.5LL	5.42	1.14	1.82	2.20	3.28	21.73	0.19	OK	0.74	0.111	OK	0.25
1.2D + 1.0SL(+z)+0.5LL	4.75	1.00	1.59	1.92	2.87	19.05	0.16	OK	0.74	0.097	OK	0.22
1.2D + 1.0SL(-z)+0.5LL	6.08	1.28	2.04	2.46	3.68	24.38	0.21	OK	0.74	0.125	OK	0.28
0.9D+1.6WL(+x)	5.63	1.19	1.89	2.28	3.41	22.58	0.19	OK	0.74	0.115	OK	0.26
0.9D+1.6WL(-x)	5.57	1.18	1.87	2.26	3.37	22.34	0.19	OK	0.74	0.114	OK	0.26
0.9D+1.6WL(+z)	7.96	1.68	2.67	3.22	4.82	31.92	0.27	OK	0.74	0.163	OK	0.37
0.9D+1.6WL(-z)	12.44	2.63	4.17	5.04	7.53	49.88	0.42	OK	0.74	0.255	OK	0.58
0.9D+SL(+x)	4.78	1.01	1.60	1.94	2.89	19.17	0.16	OK	0.74	0.098	OK	0.22
0.9D+SL(-x)	4.77	1.01	1.60	1.93	2.89	19.13	0.16	OK	0.74	0.098	OK	0.22
0.9D+SL(+z)	4.10	0.87	1.37	1.66	2.48	16.44	0.14	OK	0.74	0.084	OK	0.19
0.9D+SL(-z)	5.44	1.15	1.82	2.20	3.29	21.81	0.19	OK	0.74	0.112	OK	0.25

Load Case	kn (ϕ=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.37	0.001	315
1.4D+1.4(TL+ve)+1.7LL	0.93	0.002	630
1.4D+1.4(TL-ve)+1.7LL	0.79	0.002	630
1.2D+1.2(TL+ve)+1.6LL	0.82	0.002	630
1.2D+1.2(TL-ve)+1.6LL	0.7	0.002	630
1.2D + 1.6WL(+x)+0.5LL	0.39	0.001	315
1.2D + 1.6WL(-x)+0.5LL	0.39	0.001	315
1.2D + 1.6WL(+z)+0.5LL	0.51	0.001	315
1.2D + 1.6WL(-z)+0.5LL	0.81	0.002	630
1.2D + 1.0SL(+x)+0.5LL	0.34	0.001	315
1.2D + 1.0SL(-x)+0.5LL	0.34	0.001	315
1.2D + 1.0SL(+z)+0.5LL	0.29	0.001	315
1.2D + 1.0SL(-z)+0.5LL	0.38	0.001	315
0.9D+1.6WL(+x)	0.35	0.001	315
0.9D+1.6WL(-x)	0.35	0.001	315
0.9D+1.6WL(+z)	0.49	0.001	315
0.9D+1.6WL(-z)	0.77	0.002	630
0.9D+SL(+x)	0.3	0.001	315
0.9D+SL(-x)	0.3	0.001	315
0.9D+SL(+z)	0.25	0.001	315
0.9D+SL(-z)	0.34	0.001	315

Location	Mux (T-m)	ϕ Kn = M_{ux}/bd^2	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd.
At bottom	3.17	0.70	0.93	0.002	630.00	567.00	16ϕT	319.15	300	200
At top	0.00	0.00	0.00	0.0018	531.00	531.00	12ϕT	212.99	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :	At Bottom	**16ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c
Along Z Direction :	At Bottom	**16ϕT**	**200**	c/c	At Top	**12ϕT**	**200**	c/c

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01		
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI		

Outotec / L&T Revision No.:	0	A02	2	3	
Date	07-05-2020	05-06-2020	03-07-2020	26-10-2023	
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 03.06.20 AND REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	PRI	

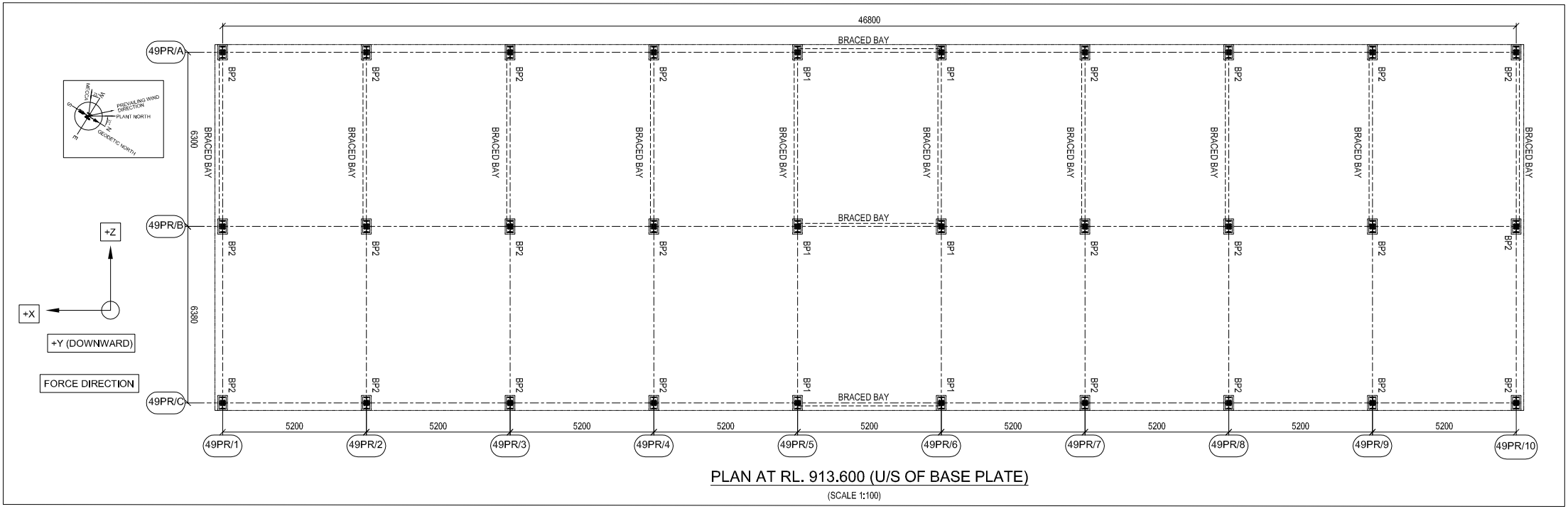
REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] For Foundation Load Data, Refer RADIATOR DESIGN CALCULATION NO. MD-216-49PR-EG-ST-CAL-2397

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	A02	2	3	
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Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI	

ATTACHMENT 1 BASE PLATE LEVEL PLAN OF RADIATOR SUPPORTING STRUCTURE



PLAN AT RL. 913.600 (U/S OF BASE PLATE)

(SCALE 1:100)